

Formal Verification of Lamport’s Mutex Algorithm for N processes

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1 Introduction

The mutual exclusion (mutex) problem, as originally introduced and solved by Dijkstra in [1], is a fundamental result in concurrent computing. Lamport’s Bakery Algorithm (LBA) is a solution to the mutex problem published in [2]. Lamport elaborates on this in [3], analysing several solution variants.

For $N \leq 5$ processes, LBA can be verified using a model checker such as TLA+, but exhaustive state expansion is not tractable for higher N . The goal of this project is to develop a formal model of LBA generalised to any finite number of concurrent processes. The model will be implemented in Lean4 to ensure that it maintains several required properties.

1.1 Problem Statement

The mutex problem is summarized by Dijkstra in [1] as follows.

To begin, consider N computers, each engaged in a process which, for our aims, can be regarded as cyclic. In each of the cycles a so-called “critical section” occurs and the computers have to be programmed in such a way that at any moment only one of these N cyclic processes is in its critical section.

A more detailed formalization of the problem was introduced by Lamport in [3], from which this proposal will take its terminology. Assuming that the i th process oscillates between critical and non-critical sections, we can represent its sequence of elementary executions as:

$$NCS_i^{[1]} \rightarrow CS_i^{[1]} \rightarrow NCS_i^{[2]} \rightarrow CS_i^{[2]} \rightarrow \dots$$

As explained in [3], there are multiple properties that must be shown to always hold to prove the correctness of the algorithm. This project will focus on the following two properties, in which $CS_i^{[k]}$ denotes the k -th execution of process i ’s critical section.

Definition 1.1 (Mutual Exclusion Property). For any pair of distinct processes i and j , no pair of operation executions $CS_i^{[k]}$ and $CS_j^{[k']}$ are concurrent.

Definition 1.2 (Deadlock Freedom Property). If there exists a looping process, then there exist an infinite number of critical section operation executions.

For a proof to be valid, these two properties must be shown to hold for every process in all state combinations. There are several variants of LBA that satisfy stronger requirements, but these are outside the scope of this project.

1.2 Formal verification in Lean4

Lean4 is a functional programming language and theorem prover with extensive mathematical tooling. Recently, it has been used to formalize and prove many problems in mathematics, including, for example, some of the Erdős problems. Currently, there are efforts in the Lean4 community to verify further problems, such as Andrew Wiles' proof of Fermat's Last Theorem. It can be used equally to ensure invariants within programs, and to prove mathematical theorems. This project will use Lean4 to implement an invariance proof that LBA ensures mutual exclusion and deadlock freedom.

2 Motivation

Formal verification is a topical field in modern Computer Science. Proving the correctness of programs is not only elegant in a theoretical sense, but is often necessary in mission-critical applications. Due to the difficulty in formalizing programs, reliability in industry is often left to automated testing. However, in specific circumstances where correctness must be ensured, formal proofs may be required.

This specific problem was chosen because it is a fundamental result in concurrent programming. Writing a formal proof for it acts as a benchmark that displays the capabilities of formal verification. Additionally, it is challenging to verify the correctness of LBA for N processes without employing an inductive approach such as this one.

Lean4 was chosen for similar reasons. Recently, it has been receiving attention for the depth of its mathematical coverage. Its library `mathlib` provides mathematical foundations and data structures that will be useful in the proof of this algorithm. The decision to choose Lean4 was also motivated by its contemporary relevance.

3 Related Works

In addition to the problem statements by Lamport and Dijkstra mentioned previously, there are several other papers that will likely be included in the literature survey. For example, in [5], Nigro proves mutual exclusion for $N \leq 5$ processes using exhaustive model checking. This result is effective for limited processes but cannot show LBA holds for larger finite numbers of processes, due to the state space expansion problem. Earlier this year, Nigro and Cicirelli published another formalization ([6]) of LBA, which uses a statistical model checking approach. This approach can scale for $N > 5$ processes, but is not as general as employing an inductive approach.

Another paper, [7], proves LBA satisfies the mutual exclusion protocol in CafeOBJ. [7] employs an approach similar to one used in this project, although its implementation is more limited. It verifies that mutual exclusion holds, but does not verify other properties such as deadlock freedom, which this project aims to accomplish. The overall system may be modelled with a node for each process, as explained by Lamport in [4].

4 Methodology

This project will use a distributed state machine with invariance proofs that ensure mutual exclusion and deadlock freedom. These properties can be maintained by creating an inductive invariant, which is proved in each state transition. The overall system may be modelled with a node for each process, as explained by Lamport in [4].

This project will implement the one-bit variant of LBA, as defined in section 3.2 of [3]. It is also noted in [8] that the ticket numbers of waiting processes in LBA have the potential to grow indeterminately large. The paper poses a solution to this issue by moving ticket numbers around a ring of length $2N$. This project will follow the same approach.

The model will consist of methods used to transition processes between states, with a validity invariant ensured after each change. It will show that for any finite N independent processes, the mutual exclusion and deadlock freedom properties will always hold across the model's processes.

4.1 Risks

The primary risk of this project is a poorly estimated scope. To mitigate against this, the project has been chosen such that it is extensible. If a model can be made that meets the previously defined requirements before the project conclusion, the scope can be adjusted to include another of the algorithms mentioned in section 3 of [3]. Alternatively, additional properties of the original algorithm may be formally verified.

Moreover, scope creep could cause the project to change focus without meeting its initial requirements. To mitigate against this, the project will strictly adhere to the previously defined methodology. Additional problems will not be included until the original requirements are met.

Other risks, such as unavailability of resources or tools, are not relevant to this project. The only tool required for the project is Lean4, which is open source, so it is improbable that it will become unavailable.

References

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